Group Project Milestone 1:
Literature Review and Exploratory Data Analysis

Forecasting Weekly Demand for Broadway Productions

A transactional analytics study of 26 years of box-office data

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National University of Singapore

IT5006 Fundamentals of Data Analytics

AY2026/2027 Semester 1

1 Literature Review

1.1 Scope and domain framing

This project analyses *Broadway weekly grosses*: 29,167 weekly box-office records covering 820 shows across 56 Manhattan theatres between August 1990 and August 2016 [19].[a] The project brief is written around e-commerce analytics, and the dataset assigned to this group sits in an adjacent domain: live entertainment. We treat the brief's requirements as domain-general, because the analytical structure is the same in both settings. Both produce a stream of transactional records at a fixed reporting grain; both raise questions of demand forecasting, revenue management and product-lifecycle risk; and both are exposed to the same two methodological hazards the brief emphasises, namely target leakage from post-outcome columns and misleading accuracy on imbalanced targets. Where the brief gives an e-commerce example we state the Broadway equivalent explicitly. The brief's warning about `customer_id` versus `customer_unique_id`, for instance, maps onto the distinction between a weekly row and a *production*: a long-running show contributes hundreds of rows, so the grouping unit for any split must be the production, not the row.

1.2 Demand and success factors for cultural products

The economics of Broadway has an established quantitative literature, and it is unusually relevant here because it studies the same institution that generated our data. Reddy et al. [14] model show success as a function of critical reviews, previews, advertising, ticket prices, show type and the timing of the opening, and find that newspaper critics exert a large and asymmetric influence. Simonoff and Ma [18] reframe success as *longevity* and fit Cox proportional-hazards models [5] to show survival, finding that show type, revival status and first-week attendance are predictive, while a Tony nomination matters far less than a Tony win. Simonoff and Cao [17] extend this with a three-way comparison of logistic regression, proportional hazards and censored linear regression on the same underlying question, confirming persistent positive effects of attendance and awards and showing that musicals outlast comparable non-musicals.

Two lessons carry into our scoping. First, the strongest predictors in this literature — reviews, awards, advertising, casting — are *not present* in our dataset, which holds only weekly box-office aggregates; any performance ceiling we report must be read against that limitation. Second, survival framing with explicit censoring is the established treatment for run length, rather than naive regression on observed duration. The wider literature agrees: Sharda and Delen [16] treat motion-picture box office as classification over discretised revenue bands rather than point regression, because the conditional distribution is wide and heavy-tailed, making a band prediction both more attainable and more actionable.

1.3 Modelling approaches for tabular panel data

For tabular data of this size the practical model space is well characterised. Regularised linear and logistic models remain the natural baseline: cheap, interpretable, and revealing of distribution shift. Tree ensembles are the usual step up — Breiman [2] bag decorrelated trees to cut variance, while gradient boosting [6] and its scalable implementations [4] fit residuals stagewise with shrinkage and explicit complexity penalties. Grinsztajn et al. [7] benchmark both families against modern deep architectures across 45 datasets and find tree-based methods still state-of-the-art at our scale, attributing the gap to inductive biases better matched to irregular tabular targets. We therefore plan a deliberately small model budget — linear/logistic baselines, a regularised variant, one tree ensemble — reused across both framings, following the brief's guidance to justify complexity with evidence rather than enumerate algorithms.

[a] **Note on the data source.** The brief's Dataset section is written around the Olist e-commerce tables, but the file distributed via Canvas is `broadway_clean.csv`. We follow its instruction to "use this exact copy rather than downloading it yourself from Kaggle or elsewhere", which exists so results stay comparable across teams. *No external data was obtained*: the Canvas file is the only source, and every target and feature here is derivable from it alone.

1.4 Feature engineering for transactional panel data

Panel data of this shape rewards a standard vocabulary: lagged target values, rolling means and dispersions, trend differences, and calendar encodings [8]. Two refinements matter here. High-cardinality categoricals such as our 56 theatres are poorly served by one-hot encoding; target encoding [12] compresses them while retaining signal, provided it is fitted inside the cross-validation loop. And the lag window must be tied to the decision it supports: a feature from week $t-1$ is unavailable to a planner committing spend four weeks out, so the admissible lag block is a function of the forecast horizon, not a free hyperparameter.

Leakage is the dominant risk. Kaufman et al. [11] formalise it as the introduction of information about the target that would not legitimately be available at prediction time, and identify exactly the two forms that threaten this dataset: post-outcome features, and improper partitioning that lets information cross the train/test boundary. On the partitioning side, Bergmeir and Benítez [1] show that standard k-fold cross-validation is invalid for serially dependent data and that evaluation must respect temporal order.

1.5 Evaluation metrics

For regression on a persistent series, absolute error against a *naive* benchmark is the honest summary; Hyndman and Koehler [9] show that many popular accuracy measures degenerate in common cases and argue for scaling errors by a naive forecast's error. We adopt the corresponding discipline of reporting a persistence baseline beside every regression result.

For imbalanced classification accuracy is actively misleading, since a constant-majority rule scores highly while achieving zero recall on the class of interest. Saito and Rehmsmeier [15] show precision–recall curves are more informative than ROC curves under strong imbalance, because ROC-AUC is insensitive to abundant true negatives. Class weighting and resampling such as SMOTE [3] are the standard remedies, though both shift the effective prior. All modelling here uses `scikit-learn` pipelines [13], which keep preprocessing fitted inside each fold.

1.6 Conclusion: trends, gaps and opportunities

Three themes emerge. The Broadway literature converges on *survival* as the natural framing for run length, and shows that type, attendance trajectory and awards carry real signal. The tabular literature converges on a small set of model families, tree ensembles being the reasonable ceiling at this scale. And the methodological literature is emphatic that on serially dependent, imbalanced data the choice of split and metric determines whether a number means anything.

The gap we can address is narrow but concrete: existing Broadway work is almost entirely retrospective, explaining what made a show succeed from variables observable only after the fact. Very little addresses the *operational* question a theatre manager faces — a short-horizon forecast of next month's demand for a show already running, from information available when the decision is made. That gap motivates the candidates in Section 2.6.

2 Exploratory Data Analysis

2.1 Dataset overview and data model

`broadway_clean.csv` contains 29,167 rows and 12 columns at a grain of one row per *week* × *show* × *theatre*.[b] The triple (`Date.Full`, `Show.Name`, `Show.Theatre`) is a verified natural key with no duplicates, and every observation is a week ending on a Sunday, the Broadway League's reporting convention. The file spans 1990-08-26 to 2016-08-14 with 1,281 distinct week-endings, 820 shows and 56 theatres, and a median of 25 productions reporting per week.

Although denormalised, the data encodes a three-level hierarchy (Figure 1): **theatres** (fixed seat count), **shows** (the creative work), and **productions** — one show's run in one theatre,

[b] **On tables and joins.** The brief asks for the tables used and joins performed. This dataset is a *single denormalised file*: no joins are required and none were performed. We meet the underlying obligation — documenting the data model — by recovering the entity structure the flat file implies (Figure 1).

the unit at which commercial decisions are taken. That distinction matters, since 27 titles appear in more than one theatre and some return after a break, so we key a production on (`show`, `theatre`, `run_block`), incrementing `run_block` after any absence beyond four weeks. This yields 942 productions overall, 902 inside the analysis window; Table 4 is the full data dictionary.

2.2 Data quality audit

The file has no nulls, no duplicate rows and no formatting faults. That apparent cleanliness is misleading: missingness has been encoded as zero, and two further defects are structural. Table 1 records every issue, its evidence and our decision; the decisions live in `src/data_load.py`, so code and table cannot drift apart. In summary: 2,309 rows (7.9%) report zero performances alongside positive attendance, which is impossible; 1,918 rows (6.6%) code gross potential as zero, including 96% of all 1996 rows; capacity is censored at 100% while 1,433 rows show gross potential above it under premium pricing; and 75 weeks are missing outright, almost all before 1993.

Critically, the zero-coded values cluster by *year* rather than by show (Figure 2b), identifying them as reporting artefacts of particular eras. Together with the thin early coverage — a handful of productions per week before 1995 against roughly 25 afterwards — this is what drives our decision to begin the analysis in 1996, leaving 28,479 rows.

Recovering an undocumented column. `Statistics.Capacity` is named as a seat count but ranges only from 10 to 100. If it is instead the percentage of seats sold, then

$$\text{seats} = \frac{\text{attendance}}{\text{performances}} \Big/ \frac{\text{capacity}}{100} \tag{1}$$

must recover a constant per theatre. It does: implied values are tight within each venue and match published seat counts to a median error of 1.7%, with the Lunt-Fontanne recovering exactly its published 1,509 seats (Figure 5). This settles the column's meaning and supplies a legitimate static feature in place of a 56-level categorical.

2.3 Temporal patterns

Total weekly gross grows roughly four-fold across the series (Figure 3), but almost none of that is demand. Between 1996 and 2015 the median ticket price rises from \$44.60 to \$93.10 (+109%) while median capacity utilisation moves only from 82% to 85% (Figure 11). A model trained on dollar levels would spend most of its capacity learning the price trend — hence our scale-free targets.

Demand is strongly seasonal, but *only at week resolution.* ISO weeks 52–53 reach a median of 90% capacity against 73% in weeks 36 and 44, a 17-point swing (Figure 6). Aggregated to months that contrast is exactly zero: December's median equals September's, because the holiday peak is a two-week phenomenon cancelled by the slow first three weeks of December. Calendar features must therefore be built at week resolution, and ours are.

Two weeks are dominated by exogenous shocks: in the week ending 16 September 2001 market gross fell from \$7.48m to \$3.47m (−54%), and during the November 2007 stagehands' strike[c] the productions able to play fell from 33 to 8 while gross collapsed from \$17.1m to \$4.6m. Neither is predictable from anything in this dataset. We retain and flag both rather than deleting inconvenient rows, and treat them as evidence that a real share of the variance here is irreducible.

2.4 Shows, theatres and lifecycles

Of the brief's three distribution analyses, two map onto this data: the *product* is the show, the *seller* is the theatre. The third cannot be done — the file records aggregate weekly attendance,

[c]**Note.** A labour dispute between the stagehands' union and the League of American Theatres and Producers closed most Broadway houses for nineteen days in November 2007. Only the handful of productions outside the affected contracts continued to play, which is why the count of reporting productions — not merely their revenue — collapses in Figure 3b.

never an individual buyer — so no customer-level problem is scopeable in Phase 2.

Show types are heavily imbalanced — 71.1% musicals, 27.8% plays, 1.1% `Special` (Figure 7) — so a classifier that always answers "Musical" scores 71% while being useless, illustrating the metric problem of Section 1.5. Recovered seat counts span roughly 500 to 1,900 and are close to independent of utilisation (Figure 8), so both carry information.

Run length is the most skewed quantity in the data: median 15 weeks, 90th percentile 68, maximum 994 (*The Phantom of the Opera* at the Majestic). Twenty-four productions were still running when the data ends and are therefore right-censored. Kaplan–Meier estimates [10] handling that censoring give median survival of 20 weeks for musicals against 13 for plays (Figure 9), reproducing the musical-longevity effect of Simonoff and Cao [17].

2.5 Relationships and the leakage they imply

Capacity is not an independent measurement but an identity over attendance, performances and seat count (Equation 1). Inverting it reconstructs capacity from same-week columns to a median absolute error of **0.48 percentage points**, so a model given same-week attendance or gross is not forecasting demand — it is reading the answer. This is the dominant leakage risk here [11], and we enforce it in code: a guard rejects any feature list containing a same-week outcome column.

2.6 Candidate problem exploration

We examined three candidates. All probes use plain logistic or linear regression inside a `scikit-learn` pipeline, deliberately untuned, on a chronological split (train $\leq$ 2013, validation 2014, test 2015 onward) that also scores the 90 test productions never seen in training.[d] A random split would be badly optimistic: *Phantom* alone contributes 994 rows, so splitting by row grades the model on interpolating a run it has memorised [1]. This is the Broadway analogue of the brief's `customer_id` warning — the unit that must not straddle the split is the production.

Candidate 1 — demand for a running production (primary). One question framed both ways: regression on capacity utilisation, and classification into three bands at training-set terciles (*Low* < 73%, *Mid*, *High* $\geq$ 89%). The stakeholder is a theatre general manager setting staffing, discounting and TKTS allocation.[e]

The horizon is a scoping decision, made with evidence (Table 2). Capacity is highly persistent — lag-1 autocorrelation 0.84 — so one week ahead a naive "same as last week" rule scores 0.728 and the model cannot beat it; but a one-week forecast is also useless, since marketing commitments cannot be changed at that notice. At four weeks persistence has decayed to 0.607 while the model holds 0.638, and the regression cuts MAE by 8.1%. We therefore scope Candidate 1 at a **four-week horizon**, where the forecast is both actionable and non-trivial.

There the classifier reaches **0.638** accuracy against a 0.315 majority and 0.607 persistence baseline, with balanced accuracy 0.637 and 0.651 on unseen productions; the regression reaches MAE 7.36 and R^2 0.546 against 8.01 and 0.399. Errors concentrate in the *Mid* band — the expected artefact of cutting a continuous quantity at tercile boundaries, not a modelling failure.

The guard's value is measurable: adding same-week attendance lifts accuracy from 0.638 to 0.875, and adding performances too to 0.902. Those are not achievements; they are what this dataset's failure mode looks like.

Candidate 2 — closure risk within eight weeks (secondary). The positive rate is 22.1%, and 208 rows are right-censored because the data ends before a closure could be observed; those

[d] **Note on splitting.** The brief recommends stratified splits for imbalanced classification. Chronology takes precedence here: a stratified random split would train on 2016 and test on 2004, and would break the production-grouping constraint. Our three-band target is balanced by construction, so stratification buys little; for the imbalanced Candidate 2 target we use class weights and time-ordered folds instead.

[e] **Note.** TKTS is the Times Square discount booth run by the Theatre Development Fund, where producers release unsold seats at a reduced price. How many to release, and how far ahead, is precisely the decision a demand forecast supports — and it is committed weeks in advance, not on the day.

are labelled missing rather than defaulted to “no”. With balanced class weights the model attains PR-AUC 0.698 against a no-skill floor of 0.204, ROC-AUC 0.887 and recall 0.806 on the minority class. Its accuracy of 0.813 barely beats the 0.796 from always predicting “stays open” — a rule catching none of the closures it exists to find, which is why PR-AUC is the honest summary [15]. This candidate contributes the imbalanced, censored target Candidate 1 lacks.

Candidate 3 — opening-month gross (deprioritised). Predicting gross in a production’s first four weeks explains only 29% of the variance in log gross, essentially all of it theatre size and seasonality. Without lags little remains, and the variables the literature calls decisive — advance sales, casting, marketing spend, reviews [14] — are absent here. We record the boundary rather than force a weak model; Table 3 scores all three against the brief’s checklist.

2.7 Key insights

1. A file with zero nulls is not a clean file: 7.9% of `Performances` and 6.6% of `Gross Potential` are missing data encoded as zero, clustered by reporting era.
2. `Statistics.Capacity` is an undocumented percentage, recoverable from the data itself to within 1.7% of published seat counts.
3. Nominal revenue growth is a price effect: ticket prices rise 109% from 1996 to 2015 while utilisation moves three points.
4. Seasonality exists only at week resolution; monthly aggregation destroys it entirely.
5. Capacity is reconstructible from same-week columns to 0.48 percentage points, making leakage the central risk and motivating a guard enforced in code.
6. Forecast difficulty is a design choice: at one week persistence solves the problem, at four weeks it is genuinely open — and four weeks is when the decision is made.

We expect a tuned Candidate 1 classifier to reach the low-to-mid 60s in accuracy — where the ceiling honestly sits, given the drivers the literature identifies are unobserved here. A result above 90% would indicate leakage, not skill.

3 Dashboard and Repository

Dashboard (Streamlit): `https://it5006-group-project-10.streamlit.app`
Repository (GitHub): `https://github.com/gengyuzhu/IT5006-Group-Project-10`
The dashboard’s five linked views read through the same `src/data_load.py` entry point as the notebooks, so its figures cannot disagree with this report. Seeds are fixed at 42, and `python src/report_stats.py` regenerates every numeric claim made here.

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A Data Model, Audit and Dictionary

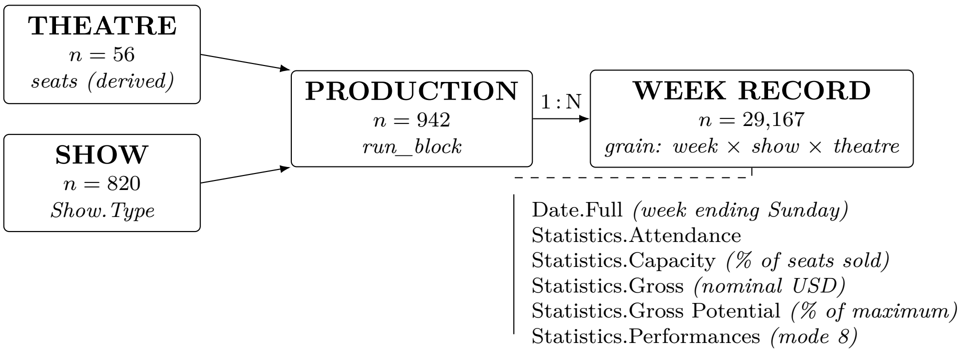

Figure 1: Entity structure recovered from the flat file. `PRODUCTION` is the decision-making unit and the grouping unit for train/test splits; seat counts are derived rather than supplied.

Table 1: Data quality audit and cleaning decisions, implemented as `CLEANING_DECISIONS` in the data-loading module.

Issue	Evidence	Decision
Coverage gaps	29 weeks missing in 1990–91 and 42 in 1991–92; 75 missing weeks in total	Restrict analysis to 1996 onward (28,479 rows); retain earlier years only as a quality exhibit
`Performances` = 0	2,309 rows (7.9%) report zero performances alongside positive attendance, which is impossible	Recode to `NaN`, flag `perf_missing`; never impute
`Gross Potential` = 0	1,918 rows (6.6%); 96% of all 1996 rows are affected, and the defect disappears entirely after 1997	Recode to `NaN`, flag `gp_missing`
`Capacity` censored	Capped at 100%, yet 1,433 rows show gross potential above 100% from premium pricing	Treat 100 as right-censored; state the censoring wherever capacity is modelled
Nominal currency	Median ticket price rises 109% between 1996 and 2015	Model scale-free measures (capacity %, gross potential %); carry `year` as a feature; no external deflator
Title-case artefacts	42 of 820 titles contain `'S` from word-wise capitalisation	Normalise to `'s`; retain the original in `show_name_raw`
Multi-theatre titles	27 titles run in more than one theatre	Key productions on (show, theatre, run_block)

Table 2: Horizon selection for Candidate 1. Persistence decays faster than the model, so the model's contribution grows with the horizon. Test period 2015-01 to 2016-08; the majority-class baseline is 0.35 at every horizon. Bold marks the better of baseline and model.

	Accuracy (3 bands)		**MAE (capacity %)**	
Horizon	Persistence	Logistic reg.	Persistence	Linear reg.
1 week	**0.728**	0.724	5.52	**5.33**
2 weeks	0.634	**0.656**	7.21	**6.52**
4 weeks	0.607	**0.638**	8.01	**7.36**

Table 3: Candidate problems against the brief's scoping checklist.

Criterion	**C1 demand (4 weeks)**	**C2 closure risk**	**C3 opening gross**
Derivable target	Yes, a column	Yes, from last observed week	Yes, weeks 1–4
No leakage	Yes, lags start at $t{-}4$	Yes, run state and lags	Yes, but no lags exist
Sufficient signal	0.638 vs 0.315 / 0.607	PR-AUC 0.698 vs 0.204	$R^2 = 0.29$ (weak)
Manageable imbalance	Balanced by construction	22% positive; weights, PR-AUC	N/A (regression)
Clear stakeholder	Theatre GM	Producer	Marketing
Verdict	**Primary**	**Secondary**	Deprioritised

B Figures

Table 4: Data dictionary for `broadway_clean.csv`. "Coded missing" records values that represent absent data rather than genuine measurements.

Column	Type	Description	Coded missing
`Date.Full`	date	Week-ending date; always a Sunday	—
`Date.Day`	int	Day component of `Date.Full`	—
`Date.Month`	int	Month component (1–12)	—
`Date.Year`	int	Year component (1990–2016)	—
`Show.Name`	text	Show title, word-wise capitalised (820 distinct)	—
`Show.Theatre`	text	Venue name (56 distinct)	—
`Show.Type`	cat	Musical / Play / Special	—
`Statistics. Attendance`	int	Total tickets sold across all performances that week	—
`Statistics. Capacity`	int	**Percentage** of available seats sold, 10–100. Not a seat count; see Equation 1	100 = censored
`Statistics. Gross`	int	Weekly box-office revenue in **nominal** USD, not inflation-adjusted	—
`Statistics. Gross Potential`	int	Percentage of maximum achievable gross; exceeds 100 under premium pricing (1,433 rows)	1,918 zeros
`Statistics. Performances`	int	Performances given that week; mode 8, the standard Broadway schedule	2,309 zeros
Derived in `src/`, not present in the raw file:			
`theatre_seats`	int	Seat count recovered via Equation 1	—
`production_id`	text	(show, theatre, run_block) — the decision unit	—
`capacity_lag`k	float	Capacity k weeks earlier, same production	gaps in run
`target_band`	cat	Low / Mid / High, at training-set terciles	—
`target_closes_soon`	bin	Closes within 8 weeks	208 censored

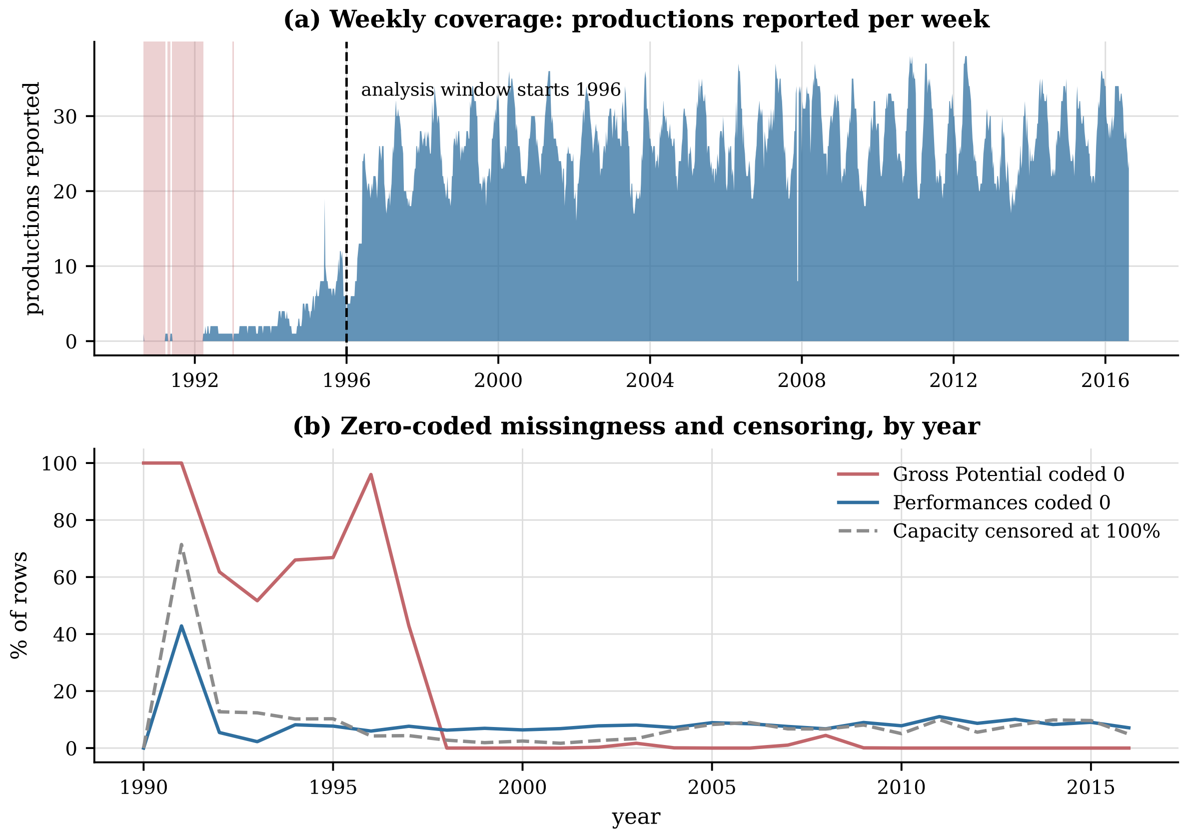

Figure 2: Data quality. (a) Productions reported per week; shaded bands are the four coverage gaps, totalling 75 missing weeks. (b) Share of rows affected by zero-coded missingness and by censoring, per year. Both defects are era effects rather than show effects, which is why the analysis window starts in 1996.

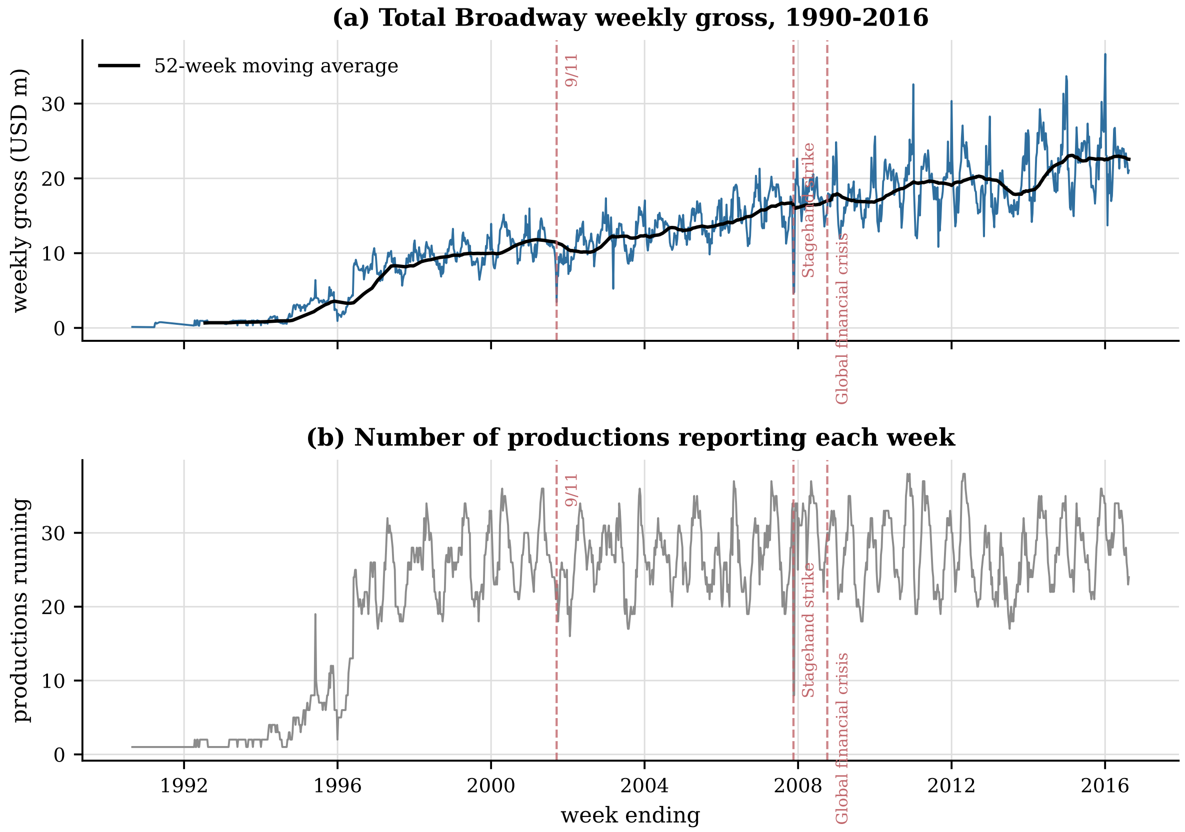

Figure 3: Total Broadway weekly gross and the number of productions reporting, 1990–2016, with the three exogenous shocks marked. The stagehands' strike is visible in both panels; the 9/11 collapse is a revenue effect without a corresponding fall in the number of productions.

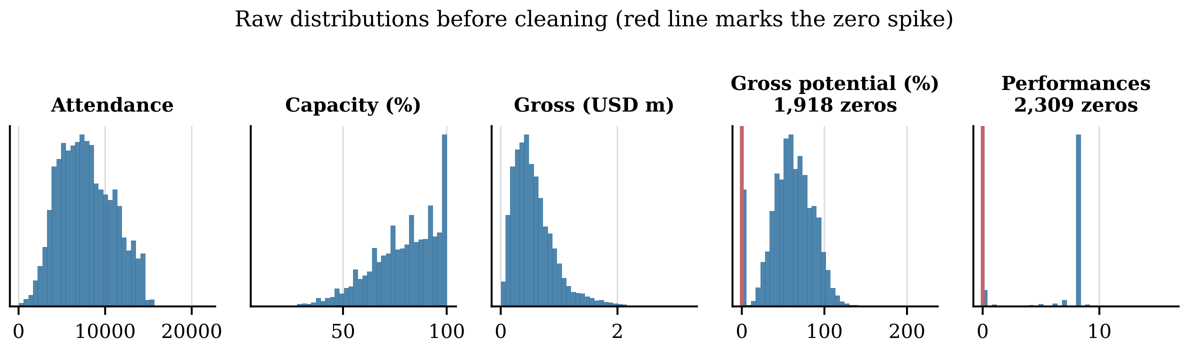

Figure 4: Raw distributions before cleaning. The red line marks the spike at zero in `Performances` and `Gross Potential`: these are missing values encoded as zero, not genuine observations.

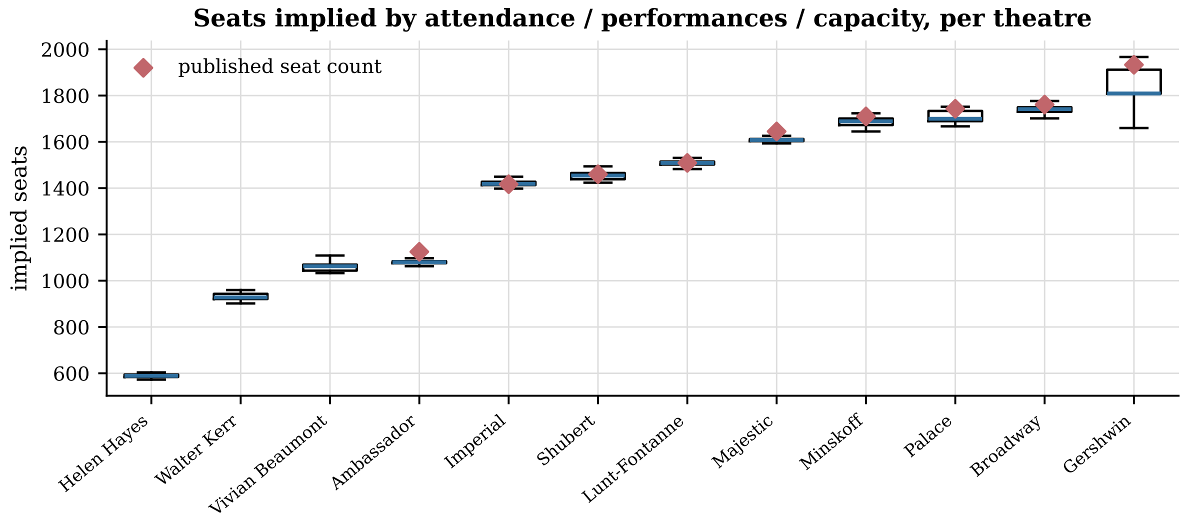

Figure 5: Validating the meaning of `Statistics.Capacity`. Boxplots show the seat count implied by attendance/performances / (capacity/100) for each theatre; diamonds are published seat counts. The implied values are tight within a theatre and match the published figures to a median error of 1.7%, establishing that the column is a percentage.

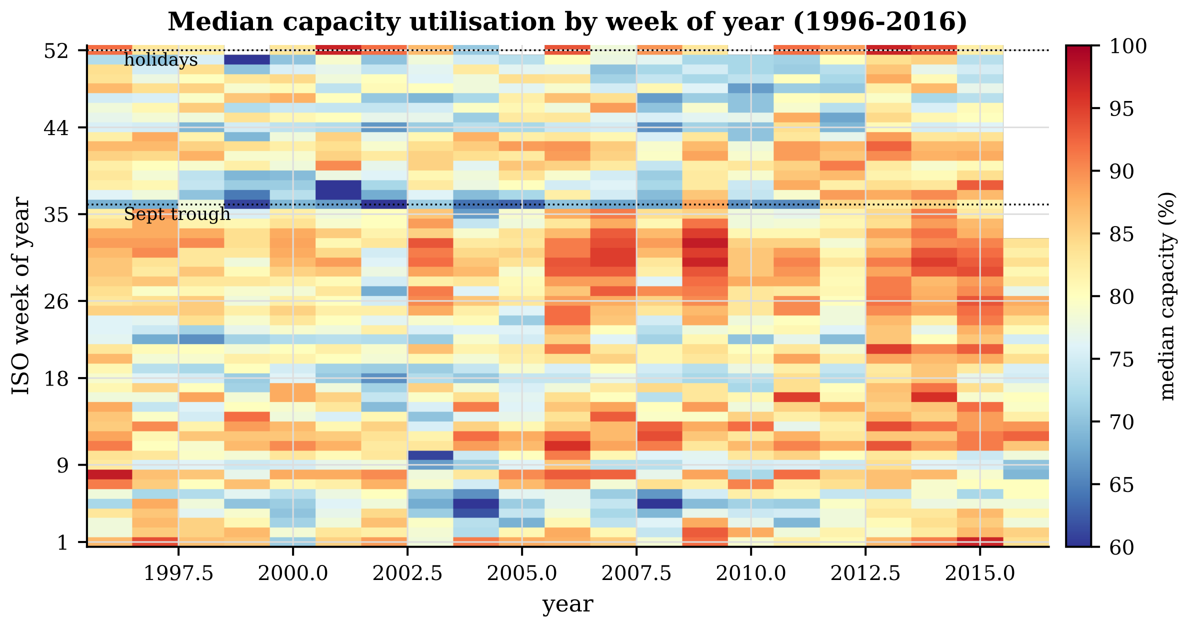

Figure 6: Median capacity utilisation by ISO week and year. The holiday band at weeks 52–53 and the September trough near week 36 persist across all twenty-one seasons.

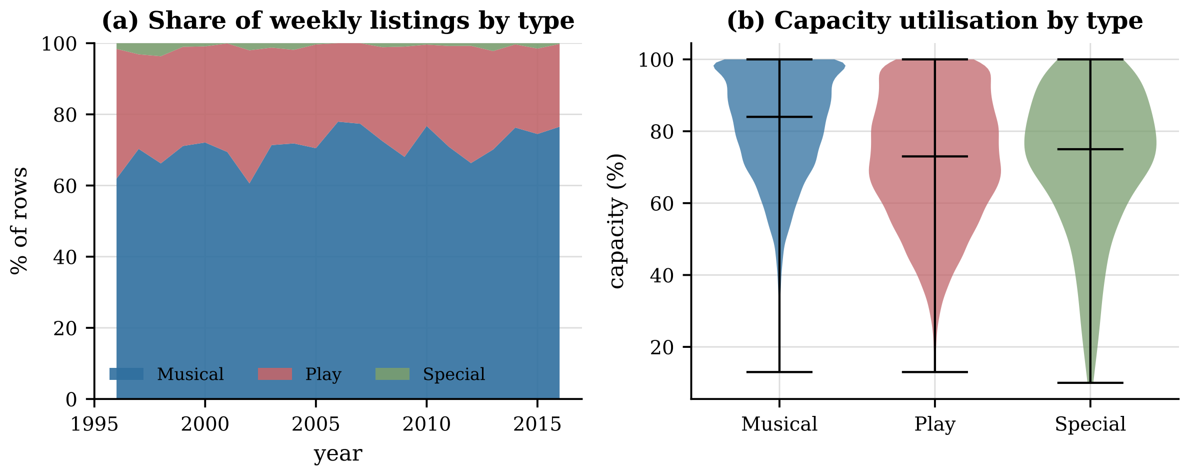

Figure 7: (a) Composition of weekly listings by show type. (b) Distribution of capacity utilisation by type. Musicals dominate the panel and sustain higher utilisation than plays.

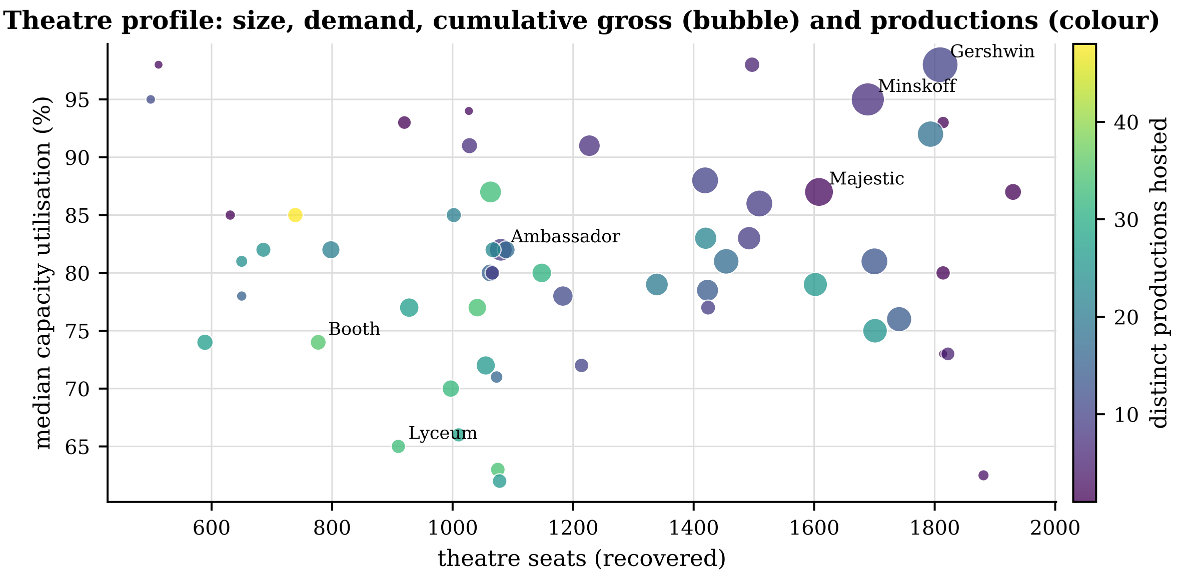

Figure 8: Theatre profile: recovered seat count against median capacity utilisation, sized by cumulative gross and coloured by the number of distinct productions hosted. Larger houses are not systematically fuller houses.

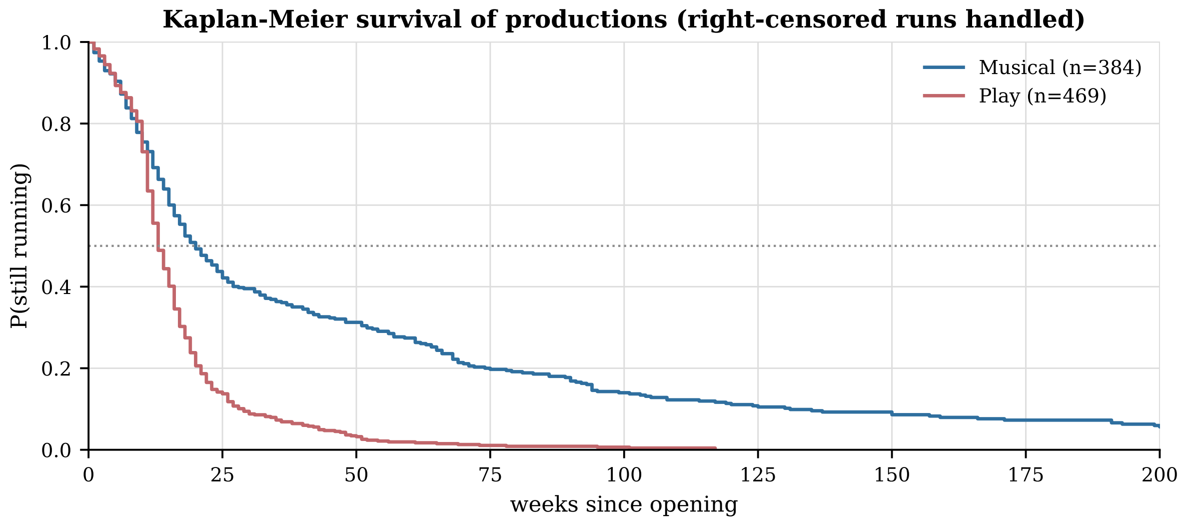

Figure 9: Kaplan–Meier survival curves for productions by show type, with right-censored runs handled explicitly. Median survival is 20 weeks for musicals and 13 weeks for plays.

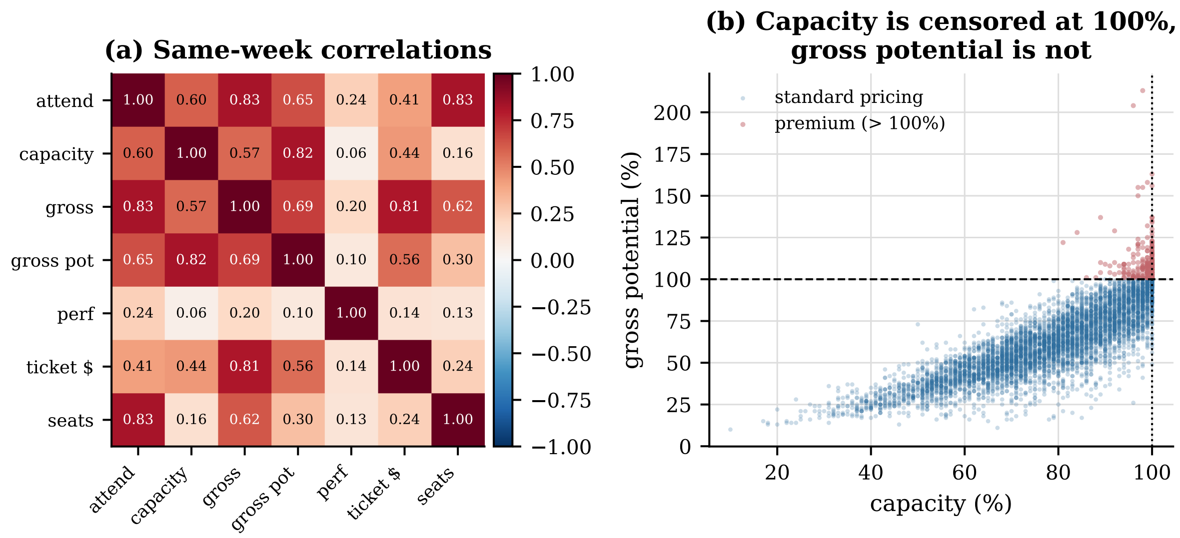

Figure 10: (a) Same-week correlations. (b) Capacity against gross potential. Capacity is censored at 100% whereas gross potential is not, so premium pricing pushes 1,433 observations above the 100% line.

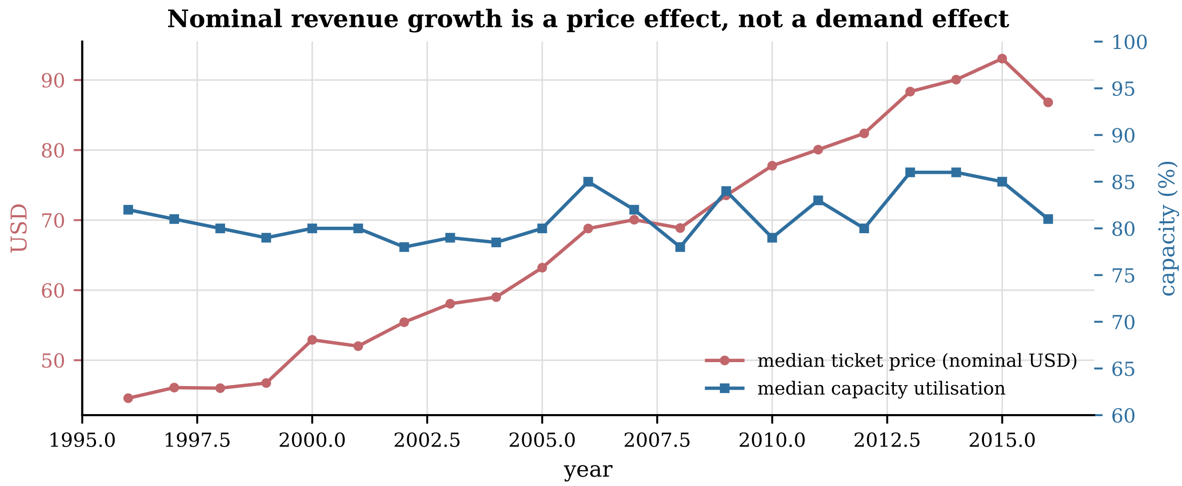

Figure 11: Median nominal ticket price against median capacity utilisation. Price rises 109% between 1996 and 2015 while utilisation moves three percentage points: growth in nominal gross is overwhelmingly a price effect.

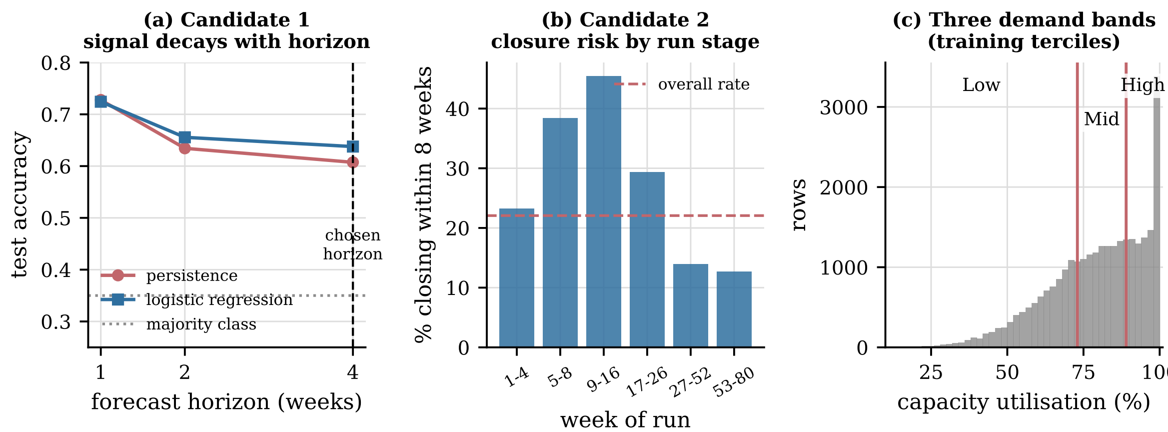

Figure 12: Candidate problem evidence. (a) Accuracy against forecast horizon: persistence decays faster than the model, so the model's contribution grows with the horizon. (b) Closure risk by stage of run. (c) The demand bands, defined by training-set terciles.